

K4H818-prog

A command-line configuration tool for the K4H818 radio module. This document is for use with K4H818-prog version 2.18.

Overview

K4H818-prog is a Python 3 program used to configure a Kits 4 Hams K4H818 radio module over its serial connection: transmit and receive frequency, CTCSS/CDCSS sub-audible tones, squelch, RX volume, and the six audio filter bypass options. It communicates with the K4H818's firmware using a simple AT+ command protocol.

On connection, K4H818-prog reads and displays the module's current configuration before making any changes, and it displays confirmation of exactly what was applied once programming completes.

Location

K4H818-prog is located:

- HamVOIP: /usr/local/sbin
- ASL3: /home/<username>

Running K4H818-prog

With no command-line options, K4H818-prog displays the current values as read from the K4H818, then runs an interactive wizard that asks for each setting in turn, showing the module's current value and accepting Enter to keep it unchanged. Before applying the new settings, it displays a full summary of the new settings and asks for confirmation.

Several command-line flags change this behavior:

Flag	Effect
(none)	Runs the normal interactive wizard, walking through every setting one at a time.
-v	Verbose mode. Prints each AT+ command as it is sent, instead of a single '+' progress character per command during programming and shows detailed diagnostic output during the initial connection to the module -- see "Connecting to the K4H818" below.
-e	Expert mode. Asks for RX analog gain (Volume1) and RX DSP gain (Volume2) directly, each 0–15, instead of the single friendly RX Volume index (0–39). Expert mode also asks for TX gain (the register 44H[11:8] nibble) and RF drive (padrv_ibit, the register 0AH[14:11] nibble) directly, each 0 to 15 -- these settings are not exposed outside expert mode and otherwise keep their current values unchanged.
-d	Defaults Mode. Presents a 5-choice menu of default configurations, instead of the interactive wizard. See "Default Configurations" below.
-r	RSSI mode. Connects to the module (same handshake as normal but requesting RSSI monitoring instead of a configuration session) and displays received signal strength once a second as it is received from the K4H818. Press Ctrl-C to stop K4H818-prog -- the K4H818 itself keeps operating normally either way, since it was already running normally to produce the RSSI stream in the first place.

Use Ctrl+C to exit the K4H818-prog at any time. Settings in the K4H818 will not be changed if a Ctrl+C exit is used.

Connecting to the K4H818

Before doing anything else, K4H818-prog hardware resets the K4H818 and waits for it to announce itself. The K4H818 sends a single "K4H818READY" line immediately after it boots, then waits up to 500ms for a response before deciding whether to enter a configuration session or proceed to normal radio operation.

K4H818-prog listens for this announcement for up to 3 seconds (allowing for reset and boot time), then responds with the appropriate request -- a configuration session for normal use, or an RSSI monitoring request for -r. If K4H818-prog reports "No K4H818READY response from the K4H818 within 3.0 seconds", the most common cause is old firmware. Update the module's firmware with K4H818-update and try again.

Running with -v shows the reset timing and every raw byte received during this process, which can help diagnose a connection issue if updating the firmware doesn't resolve it.

Ham Band Safety Check

K4H818-prog validates every transmit and receive frequency directly against the three legal amateur bands the K4H818 supports: 144–148 MHz, 222–225 MHz, and 420–450 MHz. A frequency outside all three ranges is rejected.

The frequency you choose must match your node's installed hardware low-pass filter. Selecting a frequency for the wrong band can result in no RF output power, or in out-of-spec harmonic emissions. K4H818-prog displays this warning directly above the frequency prompt.

Expert Mode (-e)

By default, RX volume is set using a single friendly index from 0 to 39, looked up against a table of paired analog/DSP gain values. Running with -e instead asks for RX analog gain (Volume1) and RX DSP gain (Volume2) directly, each ranging from 0 to 15, for finer control than the 40-step table allows.

Expert mode also asks for TX gain (the register 44H[11:8] nibble) directly, 0 to 15 -- this setting is not exposed outside expert mode, and otherwise keeps its current value unchanged.

Expert mode also asks for RF drive (0 to 15). This is set at the factory during final test and should not be changed.

Default Configurations (-d)

Running K4H818-prog with -d presents a menu of five ways to configure the module without going through the full interactive wizard:

Choice	Result
1	Resets EEPROM only. The K4H818's own built-in defaults are applied immediately. They are 435.000 MHz, CTCSS 100.0 Hz, squelch 5, RX Volume 30, all filters enabled.
2	2 meters: 146.540 MHz, CTCSS 100.0 Hz, squelch 5, RX Volume 30, all filters enabled.
3	1.25 meters: 223.400 MHz, same CTCSS/squelch/volume/filter settings as above.
4	70 cm: 446.100 MHz, same CTCSS/squelch/volume/filter settings as above.
5	Reads settings from K4H818.ini in the same folder as K4H818-prog and applies them.

K4H818.ini format

Choice 5 reads its settings from a file named K4H818.ini, which must be in the same folder as K4H818-prog itself. The file uses semicolon (;) comment lines and simple key = value pairs, with no section headers. An example file, K4H818_example.ini, documents every field with its valid range directly above it and can be copied and edited to create a working K4H818.ini file.

Field	Value
txFreq / rxFreq	Frequency in MHz (xxx.xxx). Must fall within 144.000–148.000, 222.000–225.000, or 420.000–450.000.
txcssType / rxcssType	0=none, 1=CTCSS, 2=CDCSS, 3=CTCSS with STE, 4=CDCSS with STE.
txCTCSS / rxCTCSS	CTCSS tone frequency in Hz. Used only if the matching cssType is 1 or 3.
txCDCSScode / rxCDCSScode	3-digit CDCSS code. Used only if the matching cssType is 2 or 4.
txCDCSSpolarity / rxCDCSSpolarity	N (Normal) or I (Inverted). Used only if the matching cssType is 2 or 4.
squelch	0 to 15.
rxVolume	Friendly RX Volume index, 0 to 39.
bypassPreDe, bypassVoiceHPF, bypassVoiceLPF, bypassCtcssHPF, bypassCtcssLPF, bypassRSSIlpf	Y or N. Y bypasses (disables) the filter; N keeps it enabled.
txGainNibble	Optional. 0 to 15. Defaults to 0x0B (11) if omitted. Expert-mode-only setting -- not asked in the normal wizard.

rfPower	Optional. Factory setting -- do not include unless directed.
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If K4H818.ini is missing, or any field fails validation, K4H818-prog reports the specific problem and makes no changes to the module.